Project 5: Optimization of Advanced Materials for Drones(100 Points)

Please use a new page for each part of a problem. That will help your GSI and grader!

Motivation

Material selection is one of the less glamorous but genuinely difficult problems in drone design. The constraints pull in opposite directions: structural components need to be light enough that the aircraft can actually fly usefully, but robust enough to handle the aerodynamic loads and vibration cycles that accumulate over a drone's service life — not to mention the occasional hard landing.

Composite materials are the obvious answer to the strength-to-weight problem, but they introduce their own complexity. Their effective mechanical, thermal, and electrical properties aren't just a weighted average of their constituents — they depend on particle volume fractions, phase geometry, and how efficiently stress transfers across interfaces. Getting that wrong in the design phase is expensive, which is where computational bounds like Hashin-Shtrikman become genuinely useful. Rather than committing to a material system and hoping the test data confirms it, designers can bracket the achievable property space early and narrow the experimental search considerably.

For drones specifically, this matters across more than just stiffness and density. Motors and onboard electronics generate heat that has to go somewhere; electromagnetic interference is a real concern in certain operating environments; and all of it has to be solved within a weight budget that doesn't move. Optimizing these properties in isolation is tractable — doing it simultaneously, for a real airframe, is where the computational groundwork pays off.

Overview

In this project, you will expand upon your genetic algorithm codes to find a combination of materials that achieves several objectives simultaneously. The results of this code could be used to guide early-stage experimental trials in the design of a particle-doped composite. The objective is to match a set of desired material properties while minimizing "concentration factors" that limit the effectiveness of load sharing between different phases.

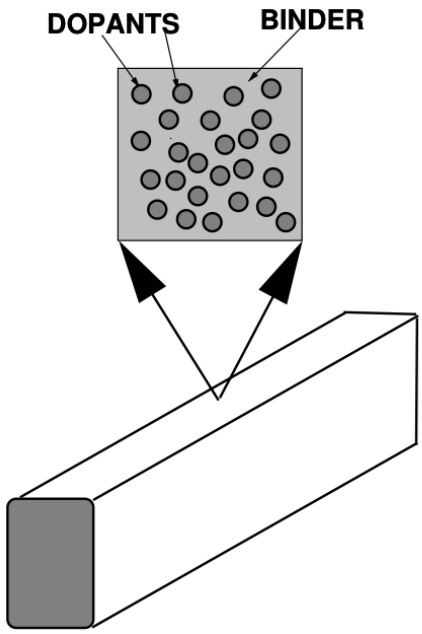


Figure 1: Example of composite material with two phases.

Effective Properties

Hashin-Shtrikman Bounds

A classical approximation for effective properties is given by the Reuss-Voigt-Wiener bounds. Improved bounds were developed in 1962 by Hashin Shtrikman based on variational principles using the concept of polarization or "filtering" of micro-macro fields. The HS bounds are the tightest possible bounds for the effective material properties of isotropic mixtures

with isotropic constituents where only the properties and volume fractions of the constituents are known. The HS bounds can estimate effective property y^* as follows:

$$y^{*,-} \leq y^* \leq y^{*,+} \tag{1}$$

In this notation:

- y is some arbitrary property (we will use several properties in this project).
- y^* represents the *effective*, or average, property in the mixture.
- $y^{*,-}$ represents the *lower bound* for this property.
- $y^{*,+}$ represents the *upper bound* for this property.

The formulas for the upper and lower bounds are not the same for all physical properties and will always be provided for you. If we know the upper and lower bounds, it must be possible to write the true effective property's value as a weighted average of the bounds:

$$y^* \approx \phi y^{*,-} + (1-\phi) y^{*,+} \stackrel{\text{def}}{=} y^{*,\phi} \tag{2}$$

For real-world applications, $0 \leq \phi \leq 1$ can be experimentally calibrated.

- If $\phi = 0$ we have the lower bound,
- If $\phi = 1$ we have the upper bound,
- If $\phi = 1/2$ we have the average of the bounds.

If no experimental data exists, selecting $\phi = 1/2$ as a starting estimate reduces the average error.

Remark: The original proofs can be found in Hashin and Shtrikman (1962). In the derivation of the HS bounds, the body is assumed to be *infinite*, the microstructure *isotropic*, and the effective responses isotropic. The bounds are the tightest possible bounds on effective material properties of mixtures using these assumptions and without microstructural information. You are *not responsible* for showing or understanding the derivation of the HS bounds but should know what physical assumptions they rely on.

In this project, we will use the Hashin and Shtrikman (HS) bounds to estimate the mechanical, thermal and electrical properties of mixtures of isotropic materials with an overall isotropic response.

Mechanical Properties

The HS bounds for the effective bulk modulus of such a mixture is given by:

$$\kappa^{*,-} \stackrel{\text{def}}{=} \kappa_1 + \frac{v_2}{\frac{1}{\kappa_2-\kappa_1} + \frac{3(1-v_2)}{3\kappa_1+4\mu_1}} \leq \kappa^* \leq \kappa_2 + \frac{1-v_2}{\frac{1}{\kappa_1-\kappa_2} + \frac{3v_2}{3\kappa_2+4\mu_2}} \stackrel{\text{def}}{=} \kappa^{*,+}, \tag{3}$$

The analogous expression for the shear modulus is:

$$\mu^{*,-} \stackrel{\text{def}}{=} \mu_1 + \frac{v_2}{\frac{1}{\mu_2-\mu_1} + \frac{6(1-v_2)(\kappa_1+2\mu_1)}{5\mu_1(3\kappa_1+4\mu_1)}} \leq \mu^* \leq \mu_2 + \frac{(1-v_2)}{\frac{1}{\mu_1-\mu_2} + \frac{6v_2(\kappa_2+2\mu_2)}{5\mu_2(3\kappa_2+4\mu_2)}} \stackrel{\text{def}}{=} \mu^{*,+}, \tag{4}$$

where κ_2 and κ_1 are the bulk moduli and μ_2 and μ_1 are the shear moduli of the respective phases (($\kappa_2 \geq \kappa_1$) and ($\mu_2 \geq \mu_1$)), and where v_2 is the second phase volume fraction. These bounds are the tightest known on isotropic effective responses, with isotropic two-phase mixtures, where only the volume fractions and material properties of the constituents are known. Use $\phi = 1/2$ to estimate the effective moduli.

Electrical Properties

We can also use the HS bounds for the overall electrical conductivity $\sigma^{c,*}$

$$\underbrace{\sigma_1^c + \frac{v_2}{\frac{1}{\sigma_2^c-\sigma_1^c} + \frac{1-v_2}{3\sigma_1^c}}}_{\sigma^{c,*,-}} \leq \sigma^{c,*} \leq \underbrace{\sigma_2^c + \frac{1-v_2}{\frac{1}{\sigma_1^c-\sigma_2^c} + \frac{v_2}{3\sigma_2^c}}}_{\sigma^{c,*,+}} \tag{5}$$

where $\sigma_2^c \geq \sigma_1^c$. Note that the superscript in $\sigma^{c,*}$ denotes "electric" and distinguishes this variable from mechanical stress. Use $\phi = 1/2$ to estimate the effective electrical conductivity.

Thermal Properties

The HS bounds for the effective thermal conductivity $\mathbb{K}^*$ of a two-phase composite are:

$$\mathbb{K}^{*,-} = \mathbb{K}_1 + \frac{v_2}{\frac{1}{\mathbb{K}_2-\mathbb{K}_1} + \frac{1-v_2}{3\mathbb{K}_1}} \leq \mathbb{K}^* \leq \mathbb{K}_2 + \frac{1-v_2}{\frac{1}{\mathbb{K}_1-\mathbb{K}_2} + \frac{v_2}{3\mathbb{K}_2}} = \mathbb{K}^{*,+} \tag{6}$$

Use $\phi = 1/2$ to estimate the effective thermal conductivity.

Concentration Factors

The average properties of a mixture are not the sole predictor of performance. If the mismatch of a property between two phases is very high, loads are not efficiently shared in the overall material, leading to locally elevated mechanical, thermal, or electrical loads. To predict the effectiveness of a mixture of materials, we need to estimate how extreme these load-concentrating factors are.

All load sharing or concentration terms rely on effective property estimates from the previous section.

Mechanical Load Sharing

The load carried by each phase in the microstructure is characterized by stress and strain concentration tensors. These provide a measure of the deviation away from the mean fields throughout the material. One can decompose averages of an arbitrary quantity over Ω into averages over the each of the phases in the following manner:

$$\langle \boldsymbol{A} \rangle_\Omega = (1/|\Omega|)\left(\int_{\Omega_1} \boldsymbol{A}\, d\Omega + \int_{\Omega_2} \boldsymbol{A}\, d\Omega \right) = v_1 \langle \boldsymbol{A} \rangle_{\Omega_1} + v_2 \langle \boldsymbol{A} \rangle_{\Omega_2} \tag{7}$$

Using this decomposition:

$$\begin{aligned} \langle \boldsymbol{\sigma} \rangle_\Omega &= v_1 \langle \boldsymbol{\sigma} \rangle_{\Omega_1} + v_2 \langle \boldsymbol{\sigma} \rangle_{\Omega_2} \\ &= v_1\, \boldsymbol{E}_1 : \langle \boldsymbol{\epsilon} \rangle_{\Omega_1} + v_2\, \boldsymbol{E}_2 : \langle \boldsymbol{\epsilon} \rangle_{\Omega_2} \\ &= \boldsymbol{E}_1 : (\langle \boldsymbol{\epsilon} \rangle_\Omega - v_2 \langle \boldsymbol{\epsilon} \rangle_{\Omega_2}) + v_2\, \boldsymbol{E}_2 : \langle \boldsymbol{\epsilon} \rangle_{\Omega_2} \\ &= \left(\boldsymbol{E}_1 + v_2 (\boldsymbol{E}_2 - \boldsymbol{E}_1) : \boldsymbol{C}^{\epsilon,2} \right) : \langle \boldsymbol{\epsilon} \rangle_\Omega, \end{aligned} \tag{8}$$

where $\boldsymbol{C}^{\epsilon,2} \stackrel{\text{def}}{=} \left(\frac{1}{v_2} (\boldsymbol{E}_2 - \boldsymbol{E}_1)^{-1} : (\boldsymbol{E}^* - \boldsymbol{E}_1) \right)$ with $\boldsymbol{C}^{\epsilon,2} : \langle \boldsymbol{\epsilon} \rangle_\Omega = \langle \boldsymbol{\epsilon} \rangle_{\Omega_2}$.

The strain concentration tensor $\boldsymbol{C}^{\epsilon,2}$ relates the average strain over phase 2 to the average strain over all phases. Similarly, for the variation in the stress we have $\boldsymbol{C}^{\epsilon,2} : \boldsymbol{E}^{*-1} : \langle \boldsymbol{\sigma} \rangle_\Omega = \boldsymbol{E}_2^{-1} : \langle \boldsymbol{\sigma} \rangle_{\Omega_2}$, which reduces to $\boldsymbol{E}_2 : \boldsymbol{C}^{\epsilon,2} : \boldsymbol{E}^{*-1} : \langle \boldsymbol{\sigma} \rangle_\Omega \stackrel{\text{def}}{=} \boldsymbol{C}^{\sigma,2} : \langle \boldsymbol{\sigma} \rangle_\Omega = \langle \boldsymbol{\sigma} \rangle_{\Omega_2}$. $\boldsymbol{C}^{\sigma,2}$ is known as the stress concentration tensor; it relates the average stress in the particle phase to that in the whole RVE. Note that once either $\boldsymbol{C}^{\epsilon,2}$ or $\boldsymbol{E}^*$ are known, the other can be determined. In the case of isotropy we may write:

$$C_\kappa^{\sigma,2} \stackrel{\text{def}}{=} \frac{1}{v_2} \frac{\kappa_2}{\kappa^*} \frac{\kappa^* - \kappa_1}{\kappa_2 - \kappa_1} \qquad \text{and} \qquad C_\mu^{\sigma,2} \stackrel{\text{def}}{=} \frac{1}{v_2} \frac{\mu_2}{\mu^*} \frac{\mu^* - \mu_1}{\mu_2 - \mu_1} \tag{9}$$

where $C_\kappa^{\sigma,2} \langle \frac{tr\boldsymbol{\sigma}}{3} \rangle_\Omega = \langle \frac{tr\boldsymbol{\sigma}}{3} \rangle_{\Omega_2}$ and where $C_\mu^{\sigma,2} \langle \boldsymbol{\sigma}' \rangle_\Omega = \langle \boldsymbol{\sigma}' \rangle_{\Omega_2}$. Clearly, the microstress fields are minimally distorted when $C_\kappa^{\sigma,2} = C_\mu^{\sigma,2} = 1$; there are no stress concentrations in a homogeneous material. For the matrix,

$$\langle \boldsymbol{\sigma} \rangle_{\Omega_1} = \frac{\langle \boldsymbol{\sigma} \rangle_\Omega - v_2 \langle \boldsymbol{\sigma} \rangle_{\Omega_2}}{v_1} = \frac{\langle \boldsymbol{\sigma} \rangle_\Omega - v_2 \boldsymbol{C}^{\sigma,2} : \langle \boldsymbol{\sigma} \rangle_\Omega}{v_1} = \frac{(\boldsymbol{1} - v_2 \boldsymbol{C}^{\sigma,2}) : \langle \boldsymbol{\sigma} \rangle_\Omega}{v_1} \stackrel{\text{def}}{=} \boldsymbol{C}^{\sigma,1} : \langle \boldsymbol{\sigma} \rangle_\Omega. \tag{10}$$

If the material is isotropic:

$$C_\kappa^{\sigma,1} \stackrel{\text{def}}{=} \frac{1}{v_1}(1 - v_2 C_\kappa^{\sigma,2}) \qquad \text{and} \qquad C_\mu^{\sigma,1} \stackrel{\text{def}}{=} \frac{1}{v_1}(1 - v_2 C_\mu^{\sigma,2}). \tag{11}$$

The fraction of the total stress carried by each phase can be determined by multiplying the concentration factors by the corresponding volume fractions:

$$\begin{aligned} \langle \boldsymbol{\sigma} \rangle_\Omega &= v_1 \langle \boldsymbol{\sigma} \rangle_{\Omega_1} + v_2 \langle \boldsymbol{\sigma} \rangle_{\Omega_2} \\ &= v_1 \boldsymbol{C}^{\sigma,1} : \langle \boldsymbol{\sigma} \rangle_\Omega + v_2 \boldsymbol{C}^{\sigma,2} : \langle \boldsymbol{\sigma} \rangle_\Omega. \end{aligned} \tag{12}$$

The strain in the matrix can be described by:

$$\langle \boldsymbol{\epsilon} \rangle_{\Omega_1} = \frac{\langle \boldsymbol{\epsilon} \rangle_{\Omega} - v_2 \langle \boldsymbol{\epsilon} \rangle_{\Omega_2}}{v_1} = \frac{\langle \boldsymbol{\epsilon} \rangle_{\Omega} - v_2 \boldsymbol{C}^{\epsilon,2} : \langle \boldsymbol{\epsilon} \rangle_{\Omega}}{v_1} = \frac{(\mathbf{1} - v_2 \boldsymbol{C}^{\epsilon,2}) : \langle \boldsymbol{\epsilon} \rangle_{\Omega}}{v_1} \stackrel{\text{def}}{=} \boldsymbol{C}^{\epsilon,1} : \langle \boldsymbol{\epsilon} \rangle_{\Omega}. \tag{13}$$

If the overall material is isotropic:

$$C_{\kappa}^{\epsilon,1} \stackrel{\text{def}}{=} \frac{1}{v_1}(1 - v_2 C_{\kappa}^{\epsilon,2}) \qquad \text{and} \qquad C_{\mu}^{\epsilon,1} \stackrel{\text{def}}{=} \frac{1}{v_1}(1 - v_2 C_{\mu}^{\epsilon,2}). \tag{14}$$

The fraction of the total strain in each phase can be determined by multiplying the concentration factors by their respective volume fractions:

$$\begin{aligned} \langle \boldsymbol{\epsilon} \rangle_{\Omega} &= v_1 \langle \boldsymbol{\epsilon} \rangle_{\Omega_1} + v_2 \langle \boldsymbol{\epsilon} \rangle_{\Omega_2} \\ &= v_1 \boldsymbol{C}^{\epsilon,1} : \langle \boldsymbol{\epsilon} \rangle_{\Omega} + v_2 \boldsymbol{C}^{\epsilon,2} : \langle \boldsymbol{\epsilon} \rangle_{\Omega}. \end{aligned} \tag{15}$$

Electrical Load Sharing

A key quantity of interest for many industrial applications is the amount of heat generated from running a current through a material, denoted H. This can be used to analyze the overall balance of energy in a system using the first law of thermodynamics:

$$\rho \dot{w} - \boldsymbol{\sigma} : \nabla \dot{\boldsymbol{u}} + \nabla \cdot q = \underbrace{H}_{\text{Joule heating source}} \tag{16}$$

In Equation 16, ρ is the mass density, w is the stored energy per unit mass, $\boldsymbol{\sigma}$ is Cauchy stress, u is the displacement field, q is heat flux, and $H = a(\mathbf{J} \cdot \mathbf{E})$, is the rate of electrical energy absorbed due to Joule heating, where $\mathbf{J}$ is the current, $\mathbf{E}$ is the electric field and $0 \leq a \leq 1$ is an absorption constant. Our objective is to determine the phase-wise load-shares of the Joule field, denoted $H = \mathbf{J} \cdot \mathbf{E}$, carried by the components in the heterogeneous mixture. The Joule heating of phase 1 is given by:

$$H_1 = \langle \boldsymbol{J} \rangle_{\Omega_1} \cdot \langle \boldsymbol{E} \rangle_{\Omega_1} = \langle \boldsymbol{J} \rangle_{\Omega_1} \cdot \boldsymbol{\sigma}_1^{c,-1} \cdot \langle \boldsymbol{J} \rangle_{\Omega_1} = \frac{\langle \boldsymbol{J} \rangle_{\Omega_1} \cdot \langle \boldsymbol{J} \rangle_{\Omega_1}}{\sigma_1^c} \tag{17}$$

The Joule heating phase 2 is given by:

$$H_2 = \langle \boldsymbol{J} \rangle_{\Omega_2} \cdot \langle \boldsymbol{E} \rangle_{\Omega_2} = \langle \boldsymbol{J} \rangle_{\Omega_2} \cdot \boldsymbol{\sigma}_2^{c,-1} \cdot \langle \boldsymbol{J} \rangle_{\Omega_2} = \frac{\langle \boldsymbol{J} \rangle_{\Omega_2} \cdot \langle \boldsymbol{J} \rangle_{\Omega_2}}{\sigma_2^c} \tag{18}$$

These can be rewritten in terms of macroscale quantities $\left(\langle \boldsymbol{J} \rangle_{\Omega}$ and $\langle \boldsymbol{E} \rangle_{\Omega}\right)$:

$$H_1 = (\boldsymbol{C}_{J,1} \cdot \langle \boldsymbol{J} \rangle_{\Omega_1}) \cdot (\boldsymbol{C}_{E,1} \cdot \langle \boldsymbol{E} \rangle_{\Omega_1}) = C_{J,1} C_{E,1} \langle \boldsymbol{J} \rangle_{\Omega_1} \cdot \langle \boldsymbol{E} \rangle_{\Omega_1} \tag{19}$$

$$H_2 = (\boldsymbol{C}_{J,2} \cdot \langle \boldsymbol{J} \rangle_{\Omega_2}) \cdot (\boldsymbol{C}_{E,2} \cdot \langle \boldsymbol{E} \rangle_{\Omega_2}) = C_{J,2} C_{E,2} \langle \boldsymbol{J} \rangle_{\Omega_2} \cdot \langle \boldsymbol{E} \rangle_{\Omega_2} \tag{20}$$

If the overall mixture is isotropic, and each of the constituents is isotropic, then the load concentration tensors become multiples of the identity matrix $\mathbf{1}$ given by:

$$\boldsymbol{C}_{E,i} = C_{E,i} \mathbf{1}$$
$$\boldsymbol{C}_{J,i} = C_{J,i} \mathbf{1}$$

For a two-phase mixture, the concentration factors are given by:

$$C_{E,1} = \frac{1}{1 - v_2} \frac{\sigma_2^c - \sigma^{c,*}}{\sigma_2^c - \sigma_1^c} \text{ and } C_{E,2} = \frac{1}{v_2} \frac{\sigma^{c,*} - \sigma_1^c}{\sigma_2^c - \sigma_1^c}, \tag{21}$$

$$C_{J,1} = \frac{\sigma_1^c}{\sigma^{c,*}(1 - v_2)} \left(\frac{\sigma_2^c - \sigma^{c,*}}{\sigma_2^c - \sigma_1^c} \right) \text{ and } C_{J,2} = \frac{\sigma_2^c}{\sigma^{c,*} v_2} \left(\frac{\sigma^{c,*} - \sigma_1^c}{\sigma_2^c - \sigma_1^c} \right). \tag{22}$$

The product of the concentration factors take the following form:

$$C_{E,1} C_{J,1} = \frac{\sigma_1^c}{\sigma^{c,*}} \left(\frac{1}{(1 - v_2)} \left(\frac{\sigma_2^c - \sigma^{c,*}}{\sigma_2^c - \sigma_1^c} \right) \right)^2 \tag{23}$$

$$C_{E,2}C_{J,2} = \frac{\sigma_2^c}{\sigma^{c,*}}\left(\frac{1}{v_2}\left(\frac{\sigma^{c,*}-\sigma_1^c}{\sigma_2^c-\sigma_1^c}\right)\right)^2. \tag{24}$$

Thermal Load Sharing

The effective thermal conductivity bounds for a two-phase mixture are given by equation 6. Recall that the conductivity of phase 2 (with volume fraction v_2) is larger than that of phase 1 ($\boldsymbol{K}_2 \geq \boldsymbol{K}_1$). Usually, v_2 corresponds to the particle material, although there can be applications where the matrix is more conductive than the particles. In that case, v_2 would correspond to the matrix material. To obtain the concentration of the thermal fields in each phase, the exact procedure holds with $\nabla\theta$ replacing $\mathbf{E}$, $\mathbf{q}$ replacing $\mathbf{J}$, and $\boldsymbol{K}$ replacing $\sigma^{\mathbf{c}}$. For thermal properties, use the following concentration tensors:

$$\boldsymbol{C}_{\theta,2}\cdot\langle\nabla\theta\rangle_\Omega = \langle\nabla\theta\rangle_{\Omega_2} \text{ where } \boldsymbol{C}_{\theta,2} = \frac{1}{v_2}(\boldsymbol{K}_2-\boldsymbol{K}_1)^{-1}\cdot(\boldsymbol{K}^*-\boldsymbol{K}_1) \tag{25}$$

$$\boldsymbol{C}_{\theta,1}\cdot\langle\nabla\theta\rangle_\Omega = \langle\nabla\theta\rangle_{\Omega_1} \text{ where } \boldsymbol{C}_{\theta,1} = \frac{1}{v_1}(1-v_2\boldsymbol{C}_{\theta,2}) = \frac{1-v_2\boldsymbol{C}_{\theta,2}}{1-v_2} \tag{26}$$

$$\boldsymbol{C}_{q,2}\cdot\langle\boldsymbol{q}\rangle_\Omega = \langle\boldsymbol{q}\rangle_{\Omega_2} \text{ where } \boldsymbol{C}_{q,2} = \boldsymbol{K}_2\cdot\boldsymbol{C}_{\theta,2}\cdot\boldsymbol{K}^{*-1} \tag{27}$$

$$\boldsymbol{C}_{q,1}\cdot\langle\boldsymbol{q}\rangle_\Omega = \langle\boldsymbol{q}\rangle_{\Omega_1} \text{ where } \boldsymbol{C}_{q,1} = \frac{\mathbf{1}-v_2\boldsymbol{C}_{q,2}}{1-v_2} \tag{28}$$

Optimization

In the following sections, we will describe our goals for the mechanical, electrical, and thermal properties of the material, then combine those goals into a single scalar value to be minimized by the genetic algorithm. In general, all the weights used could be distinct. For this problem, all weights will be identical for simplicity. All necessary values are defined in the variable glossary.

Optimization of Mechanical Properties

Our goal for the mechanical properties is to computationally design the macroscale effective bulk and shear moduli κ^* and μ^*. Recall that the effective properties can be estimated using convex combinations of the Hashin-Shtrikman bounds as approximations for the effective moduli $\kappa^* \approx \phi\kappa^{*,+} + (1-\phi)\kappa^{*,-}$ and $\mu^* \approx \phi\mu^{*,+} + (1-\phi)\mu^{*,-}$, where $0 \leq \phi \leq 1$ — for this project, we will use $\phi = 1/2$. The micro-macro objective function is:

$$\boxed{\Pi^{mechanical} = w_1|\frac{\kappa^{*,Des} - \kappa^*}{\kappa^{*,Des}}|^2 + w_1|\frac{\mu^{*,Des} - \mu^*}{\mu^{*,Des}}|^2 + \hat{w}_3|\frac{C_\kappa^{\sigma,2} - TOL_\kappa}{TOL_\kappa}|^2 + \hat{w}_4|\frac{C_\mu^{\sigma,2} - TOL_\mu}{TOL_\mu}|^2 \\ +\hat{w}_5|\frac{C_\kappa^{\sigma,1} - TOL_\kappa}{TOL_\kappa}|^2 + \hat{w}_6|\frac{C_\mu^{\sigma,1} - TOL_\mu}{TOL_\mu}|^2}$$

Any weight of the form w_i represents a scalar constant. Any weight of the form $\hat{w}_i$ represents a unilateral constraint, defined as a piecewise constant function or "step function."

$$\hat{w}_3 = \begin{cases} w_3 & C_\kappa^{\sigma,2} > TOL_\kappa \\ 0 & \text{otherwise} \end{cases} \tag{29}$$

$$\hat{w}_4 = \begin{cases} w_4 & C_\mu^{\sigma,2} > TOL_\mu \\ 0 & \text{otherwise} \end{cases} \tag{30}$$

$$\hat{w}_5 = \begin{cases} w_5 & C_\kappa^{\sigma,1} > TOL_\kappa \\ 0 & \text{otherwise} \end{cases} \tag{31}$$

$$\hat{w}_6 = \begin{cases} w_6 & C_\mu^{\sigma,1} > TOL_\mu \\ 0 & \text{otherwise} \end{cases} \tag{32}$$

In general, the weights on the μ and κ terms could be different. They are set to be the same here to prevent confusion with the weights for the step functions in the other components of the overall cost function.

Here the design variables are $\mathbf{\Lambda} = \{\kappa_2, \mu_2, v_2\}$, and their constrained ranges are: $\kappa_2^{(-)} \leq \kappa_2 \leq \kappa_2^{(+)}$, $\mu_2^{(-)} \leq \mu_2 \leq \mu_2^{(+)}$ and $v_2^{(-)} \leq v_2 \leq v_2^{(+)}$. All values are specified in the glossary on the first page of this assignment.

Optimization of Electrical Properties

In order to utilize the previous results, we now provide examples of how to optimize the material properties. In particular, we will optimize the overall electrical conductivity of a mixture of material, subject to Joule heating constraints on each phase. The effective conductivity is given by $\sigma^{c,*}$ using convex combinations of the HS bounds as approximations for the effective moduli $\sigma^{c,*} \approx \phi\sigma^{c,*,+} + (1-\phi)\sigma^{c,*,-}$, where $0 \leq \phi \leq 1$ — for this project, we will use $\phi = 1/2$. The micro-macro objective function is:

$$\boxed{\Pi^{elec} = w_1|\frac{\sigma^{c,*,Des} - \sigma^{c,*}}{\sigma^{c,*,Des}}|^2 + \hat{w}_2|\frac{C_{J1}C_{E1} - TOL_\sigma}{TOL_\sigma}|^2 + \hat{w}_3|\frac{C_{J2}C_{E2} - TOL_\sigma}{TOL_\sigma}|^2,} \tag{33}$$

where:

$$\hat{w}_2 = \begin{cases} w_2 & \frac{C_{J1}C_{E1} - TOL_\sigma}{TOL_\sigma} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{34}$$

$$\hat{w}_3 = \begin{cases} w_3 & \frac{C_{J2}C_{E2} - TOL_\sigma}{TOL_\sigma} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{35}$$

Here the design variables are $\mathbf{\Lambda} = \{\sigma_{e,2}, v_2\}$, and their constrained ranges are:

- $(\sigma_2^c)^{(-)} \leq \sigma_2^c \leq (\sigma_2^c)^{(+)}$ and
- $v_2^{(-)} \leq v_2 \leq v_2^{(+)}$.

Optimization of Thermal Properties

In order to utilize the previous results, we now provide examples of how to optimize the material properties. In particular, we will optimize the overall thermal conductivity of a multi-phase mixture, subject to thermal constraints on each phase. The effective thermal conductivity is given by $\boldsymbol{K}^*$ using convex combinations of the HS bounds as approximations for the effective property: $\boldsymbol{K}^* \approx \phi \boldsymbol{K}^{*+} + (1-\phi)\boldsymbol{K}^{*-}$, where $0 \leq \phi \leq 1$ — for this project, we will use $\phi = 1/2$. The micro-macro objective function is:

$$\boxed{\Pi^{thermal} = w_1 |\frac{\boldsymbol{K}^{*,Des} - \boldsymbol{K}^*}{\boldsymbol{K}^{*,Des}}|^2 + \hat{w}_2 |\frac{C_{q1} - TOL_{\boldsymbol{K}}}{TOL_{\boldsymbol{K}}}|^2 + \hat{w}_3 |\frac{C_{q2} - TOL_{\boldsymbol{K}}}{TOL_{\boldsymbol{K}}}|^2} \tag{36}$$

$$\hat{w}_2 = \begin{cases} w_2 & \frac{C_{q1} - TOL_{\boldsymbol{K}}}{TOL_{\boldsymbol{K}}} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{37}$$

$$\hat{w}_3 = \begin{cases} w_3 & \frac{C_{q2} - TOL_{\boldsymbol{K}}}{TOL_{\boldsymbol{K}}} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{38}$$

Here the design variables are $\mathbf{\Lambda} = \{\boldsymbol{K}_2, v_2\}$, and their constrained ranges are

- $(\boldsymbol{K}_2)^{(-)} \leq \boldsymbol{K}_2 \leq (\boldsymbol{K}_2)^{(+)}$ and
- $v_2^{(-)} \leq v_2 \leq v_2^{(+)}$.

Multi-Property Optimization

We will now combine the cost functions for each material property to optimize all properties simultaneously. The micro-macro objective function is:

$$\boxed{\Pi^{total} = W_1 \Pi^{electrical} + W_2 \Pi^{thermal} + W_3 \Pi^{mechanical}} \tag{39}$$

and say our design strings will be of the form

$$\mathbf{\Lambda}^i = \{\kappa_2, \mu_2, \sigma_2^c, \boldsymbol{K}_2, v_2\}^i, \quad \text{for } i = 1, ..., \mathsf{S}. \tag{40}$$

There are two characteristics of such a formulation which make the application of gradient-based minimization schemes, such as Newton's method, difficult:

- (I) the incorporation of limits on the microfield behavior, as well as design search space restrictions, renders the objective function not continuously differentiable in design space.
- (II) the objective function is nonconvex, i.e. the system Hessian is not positive definite (invertible) throughout design space.

IMPLEMENTATION

We have provided the skeleton code in a Python Jupyter notebook, Hashin-Shtrikman_INCOMPLETE.ipynb. Follow the steps outlined here and in the notebook.

Steps

0. Fill in the parameters in the first cell of the notebook.
1. Fill in values from the design string as ordered in Eqn. 40.
2. Compute property bounds from Eqns. 5, 6, 3, and 4.
3. Compute effective properties via Eqn. 2.
4. Determine concentration tensors: Eqns. 23, 24, 25, 26, 27, 28, 9, and 11.
5. Compute the weights in each material cost function: Eqns 34, 35, 37, 38, 29, 30, 31, and 32.
6. Compute cost functions of each material property 33, 36, 29, and the total cost 39.
7. For Case B, make the necessary modifications in the offspring process.
8. In Case C, define constants a and b to allow for mutations in offspring.
9. For Case C, make the necessary modifications in the offspring process.
10. Run the code and write your report.

Project Parameters

Table 1: Project Parameters

Variable	Value	Units	Description
κ_1	90	GPa	phase 1 bulk modulus
μ_1	20	GPa	phase 1 shear modulus
σ_1^c	1.1×10^7	S/m	phase 1 electrical conductivity
$\mathbb{K}_1$	4.5	W/m/K	phase 1 thermal conductivity
$\kappa^{*,D}$	125	GPa	desired effective bulk modulus
$\mu^{*,D}$	50	GPa	desired effective shear modulus
$\sigma^{c,*,D}$	2.2×10^7	S/m	desired effective electrical conductivity
$\mathbb{K}^{*,D}$	7.5	W/m/K	desired effective thermal conductivity
TOL_κ	0.5	—	bulk modulus tolerance
TOL_μ	0.5	—	shear modulus tolerance
$TOL_{\mathbb{K}}$	0.5	—	thermal tolerance
TOL_σ	0.8	—	electrical tolerance
κ_2^-, κ_2^+	$\{\kappa_1, 10\kappa_1\}$	GPa	bulk modulus search bounds
μ_2^-, μ_2^+	$\{\mu_1, 10\mu_1\}$	GPa	shear modulus search bounds
$\mathbb{K}_2^-, \mathbb{K}_2^+$	$\{\mathbb{K}_1, 10\mathbb{K}_1\}$	W/m/K	thermal conductivity search bounds
$\sigma_2^{c,-}, \sigma_2^{c,+}$	$\{\sigma_1^c, 10\sigma_1^c\}$	S/m	electrical conductivity search bounds
v_2^-, v_2^+	$\{0, \frac{2}{3}\}$	—	second phase volume fraction search bounds
W_1, W_2, W_3	$\frac{1}{3}$	—	total cost function weights
w_1	1	—	material property matching weights
$w_j, j = 2, 3, 4, 5, 6$	0.5	—	concentration factor weights
γ	0.5	—	Hashin-Shtrikman averaging constant
K	10	—	number of children for GA
P	10	—	number of parents for GA
S	200	—	number of design strings in a population
G	5000	—	max number of generations

Deliverables

Use a genetic algorithm to find the optimal values for the second phase material parameters as well as the optimal volume fraction required to achieve desired material properties.

Consider the following 3 cases:

- **Case A:** Keep the 10 parents after each generation, produce 10 offspring, and generate 180 new random strings
- **Case B:** Do *not* keep the 10 parents after each generation, produce 10 offspring, and generate 190 new random strings
- **Case C:** Keep the 10 parents after each generation, produce 10 offspring *with mutations*, and generate 180 new random strings

To produce offspring with mutations, instead of using combinations of parents with random weights $0 \leq \phi \leq 1$, use $-1/2 \leq \phi \leq 3/2$ instead.

1. **(40 Points)** Run the genetic algorithm for 5000 generations. Due to the random nature of the initial parameters, you may have to run the code several times to get good convergence. There is no required tolerance. Provide convergence plots showing the cost of the best design for each generation and the mean cost of the top 10 designs. Plots should be appropriately labelled and captioned in your report (3 plots total, 1 for each case). **These plots should be on a log-log scale to be intelligible. Do NOT plot the average cost for entire generations**.
2. **(40 Points)** Discuss the performance of each GA variation (Case A, Case B, and Case C). Refer to specific result as evidence to compare how the parameters directly affected the convergence rate, the repeatability, the accuracy, etc. Explain in depth why each case yielded these results.
3. **(20 Points)** Report your best-performing 4 designs for all 3 Cases into 3 separate tables similar to the following. **Round data to 4 digits. Do not copy and paste 16-digit numbers.**

Design	Λ_1	Λ_2	Λ_3	etc	Π
1					
2					
3					
4					

Table 2: The top 4 system parameter performers.

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Submission

Only submit the pdf of this report to gradescope before 11:59 pm on the due date.